

IMAGES IN PYTHON

Nada Fitrieyatul Hikmah

Loading Images (DICOM)

- **imageio** : read and save images.

To install imageio, use one of the following methods:

- If you are in a conda env: `conda install -c conda-forge imageio`
- If you have pip: `pip install imageio`
- Good old `python setup.py install`

```
In [1]: 1 import imageio
        2
        3 im = imageio.imread('D:\Teaching\Pengolahan Citra Medika\Database\IM-0001-0001.dcm')
        4 type(im)
```

```
Out[1]: imageio.core.util.Array
```

```
In [2]: 1 im
```

```
Out[2]: Array([[ -991, -1011, -1024, ..., -975, -955, -966],
               [ -998, -1007, -1024, ..., -971, -966, -977],
               [-1000, -1005, -1021, ..., -974, -975, -968],
               ...,
               [ -833,  -804,  -753, ..., -331, -377, -386],
               [ -912,  -885,  -846, ..., -540, -584, -591],
               [ -947,  -939,  -934, ..., -685, -723, -736]], dtype=int16)
```

Metadata

- Images are always acquired in a specific context. This information is often referred to as **metadata**.
- Accessible in image objects through the **meta** dictionary attribute.

```
In [3]: 1 im.meta
```

```
Out[3]: Dict([('TransferSyntaxUID', '1.2.840.10008.1.2.1'),  
             ('SOPClassUID', '1.2.840.10008.5.1.4.1.1.2'),  
             ('SOPInstanceUID',  
              '1.3.12.2.1107.5.1.4.50025.30000005060811542834300000777'),  
             ('StudyDate', '20050601'),  
             ('SeriesDate', '20050601'),  
             ('AcquisitionDate', '20050608'),  
             ('ContentDate', '20050608'),  
             ('StudyTime', '120000.000000 '),  
             ('SeriesTime', '120000.000000 '),  
             ('AcquisitionTime', '103434.945159 '),  
             ('ContentTime', '103434.945159 '),  
             ('Modality', 'CT'),
```

```
In [4]: 1 im.meta['Modality']
```

```
Out[4]: 'CT'
```

```
In [5]: 1 im.meta.keys()
```

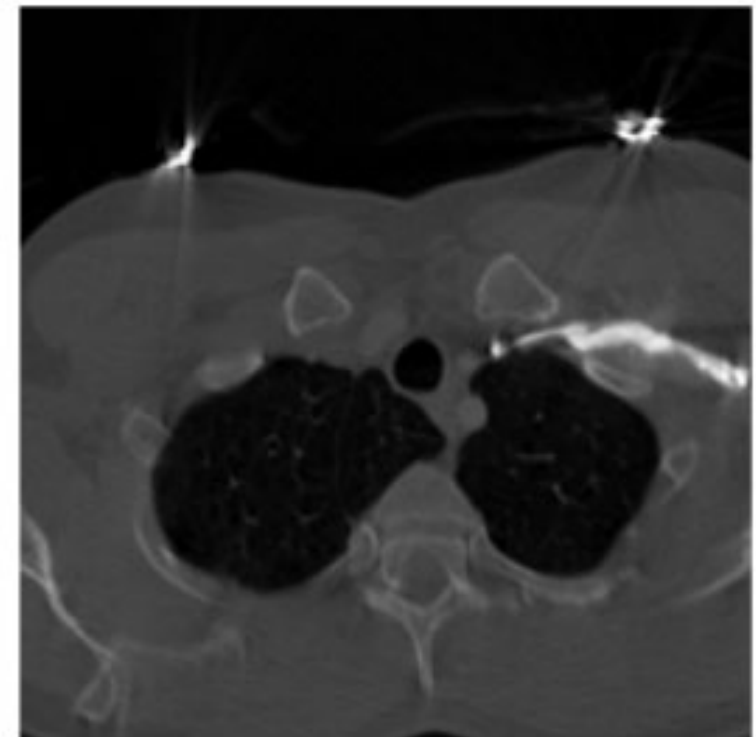
```
Out[5]: odict_keys(['TransferSyntaxUID', 'SOPClassUID', 'SOPInstanceUID', 'StudyDate',  
                   'SeriesDate', 'AcquisitionDate', 'ContentDate', 'StudyTime', 'SeriesTime', 'Acq  
u  
isitionTime', 'ContentTime', 'Modality', 'Manufacturer', 'InstitutionName', 'S  
tudyDescription', 'SeriesDescription', 'PatientName', 'PatientID', 'PatientBirt  
hDate', 'PatientSex', 'PatientAge', 'StudyInstanceUID', 'SeriesInstanceUID', 'S  
eriesNumber', 'AcquisitionNumber', 'InstanceNumber', 'ImagePositionPatient', 'I  
mageOrientationPatient', 'SamplesPerPixel', 'Rows', 'Columns', 'PixelSpacing',  
'BitsAllocated', 'BitsStored', 'HighBit', 'PixelRepresentation', 'RescaleInterc  
ept', 'RescaleSlope', 'PixelData', 'shape', 'sampling'])
```

Plotting Images

- Matplotlib's `imshow()` function displays 2D image data.
- Many colormaps available but often shown in grayscale (`cmap='gray'`)
- Axis ticks and labels are often not useful for images

In [12]:

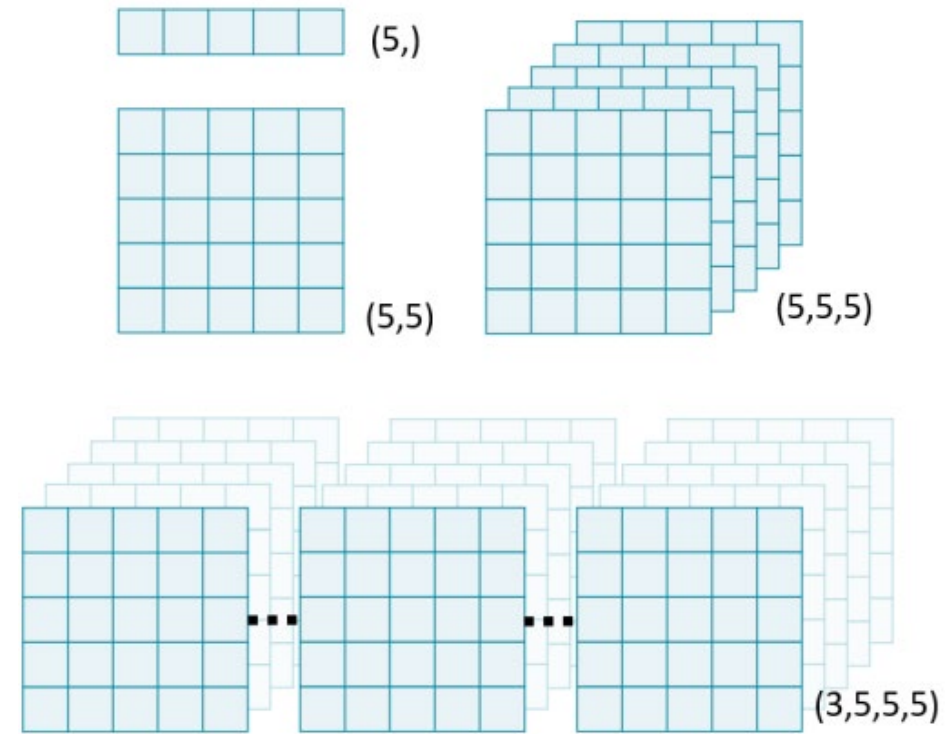
```
1 import matplotlib.pyplot as plt
2 plt.imshow(im,cmap='gray')
3 plt.axis('off')
4 plt.show()
```



A vertical bar on the left side of the slide, transitioning from orange at the top to purple at the bottom.

N-DIMENSIONAL IMAGES

N-dimensional images are stacks of arrays



```
In [30]: 1 import imageio
          2 import numpy as np
          3 im1=imageio.imread('D:\Teaching\Pengolahan Citra Medika\Database\dataset1_Thorax_Abdomen\IM-0001-0015.dcm')
          4 im2=imageio.imread('D:\Teaching\Pengolahan Citra Medika\Database\dataset1_Thorax_Abdomen\IM-0001-0016.dcm')
          5 im3=imageio.imread('D:\Teaching\Pengolahan Citra Medika\Database\dataset1_Thorax_Abdomen\IM-0001-0017.dcm')
          6
          7 im1.shape
```

Out[30]: (512, 512)

```
In [51]: 1 vol = np.dstack([im1,im2,im3])
          2 vol.shape
```

Out[51]: (512, 512, 3)

Loading volumes directly

Imageio.volread():

- Read multi-dimensional data directly.
- Assemble a volume from multiple images

```
1 import os
2 os.listdir('D:\\Teaching\\Pengolahan Citra Medika\\Databa
3 'IM-0001-0016.dcm',
4 'IM-0001-0017.dcm',
5 'IM-0001-0018.dcm',
6 'IM-0001-0019.dcm',
7 'IM-0001-0020.dcm',
8 'IM-0001-0021.dcm',
9 'IM-0001-0022.dcm',
10 'IM-0001-0023.dcm',
11 'IM-0001-0024.dcm',
12 'IM-0001-0025.dcm',
13 'IM-0001-0026.dcm',
14 'IM-0001-0027.dcm',
15 'IM-0001-0028.dcm',
16 'IM-0001-0029.dcm',
17 'IM-0001-0030.dcm',
18 'IM-0001-0031.dcm',
19 'IM-0001-0032.dcm',
20 'IM-0001-0033.dcm',
21 'IM-0001-0034.dcm',
22 'IM-0001-0035.dcm'.
```

In [60]:

```
1 import imageio
2 vol1 = imageio.volread('D:\\Teaching\\Pengolahan Citra Medika\\Database\\dataset1_Thorax_Abdomen')
3 vol1.shape
```

Reading DICOM (examining files): 1/291 files (0.392/291 files (31.6189/291 files (64.9%233/291 files (80.1%291/291 files (100.0%)

Found 1 correct series.

Reading DICOM (loading data): 37/291 (12.7%79/291 (27.1%121/291 (41.6161/291 (55.3205/291 (70.4240/291 (82.5274/291 (94.2291/291 (100.0%)

Out[60]: (291, 512, 512)

Sampling and field of view

- **Sampling rate:** physical space covered by each element.
- **Field of view:** physical space covered along each axis.

```
1 #sampling rate (in mm)
2 d0, d1, d2 = vol1.meta['sampling']
3 d0, d1, d2 #d0 = jarak antar slicing,
```

(2.0, 0.5078125, 0.5078125)

```
1 #Field of view (in mm)
2 n0 * d0, n1 * d1, n2 * d2
```

(582.0, 260.0, 260.0)

A vertical bar on the left side of the slide, transitioning from orange at the top to purple at the bottom.

IMAGE PLOTTING

Plotting multiple images at once

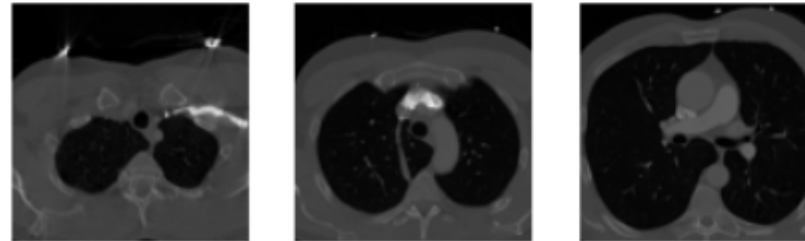
- `plt.subplots`: creates a figure canvas with multiple **AxesSubplots** objects.

```
1 import imageio
2 vol2 = imageio.volread('D:\Teaching\Pengolahan Citra Medika\Database\dataset
3 fig, axes = plt.subplots(nrows=1,ncols=3)
4
5 axes[0].imshow(vol[0],cmap='gray')
6 axes[1].imshow(vol[20],cmap='gray')
7 axes[2].imshow(vol[50],cmap='gray')
8
9 for ax in axes:
10     ax.axis('off')
11
12 plt.show()
```

Reading DICOM (examining files): 1/291 files (0.3240/291 files (82.5%291/291 files (100.0%))

Found 1 correct series.

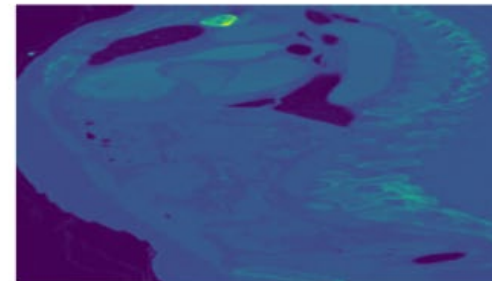
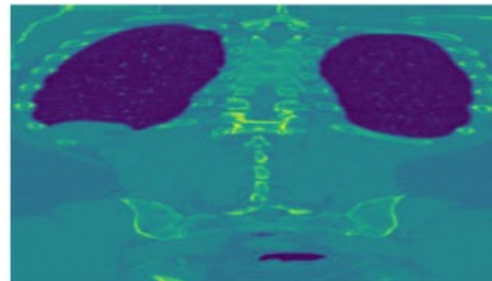
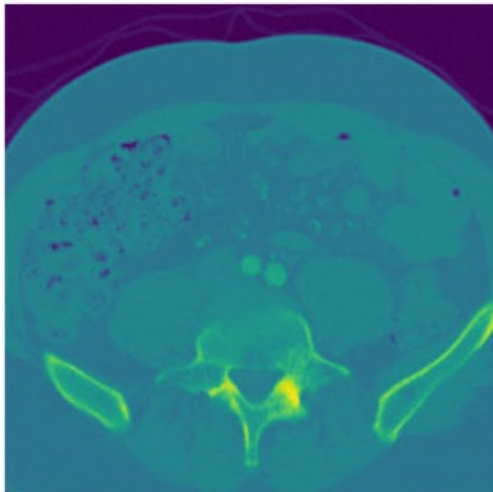
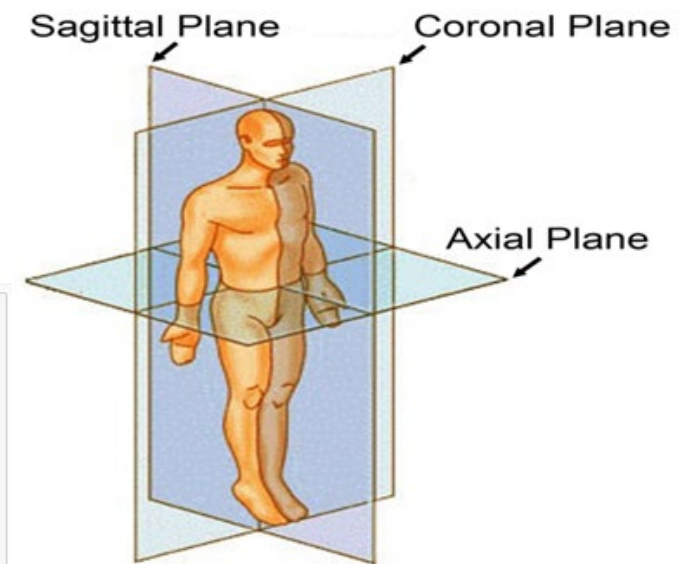
Reading DICOM (loading data): 96/291 (33.0%200/291 (68.7291/291 (100.0%))



Another views

```
1 import imageio
2 vol3 = imageio.volread('D:\Teaching\Pengolahan Citra Medika\Database\dataset1_Thorax_Abdomen')
3 fig, axes = plt.subplots(nrows=1,ncols=3, figsize=(20,10))
4 view_axial = vol3[200, :, :]
5 view_coronal = vol3[:, 450, :]
6 view_sagittal = vol3[:, :, 300]
7
8 axes[0].imshow(view_axial)
9 axes[1].imshow(view_coronal)
10 axes[2].imshow(view_sagittal)
11
12 for ax in axes:
13     ax.axis('off')
14
15 plt.show()
16
```

Reading DICOM (examining files): 1/291 files (0.3226/291 files (77.7%291/291 files (100.0%))
Found 1 correct series.
Reading DICOM (loading data): 97/291 (33.3%190/291 (65.3286/291 (98.3291/291 (100.0%))



Modifying the aspect ratio (1)

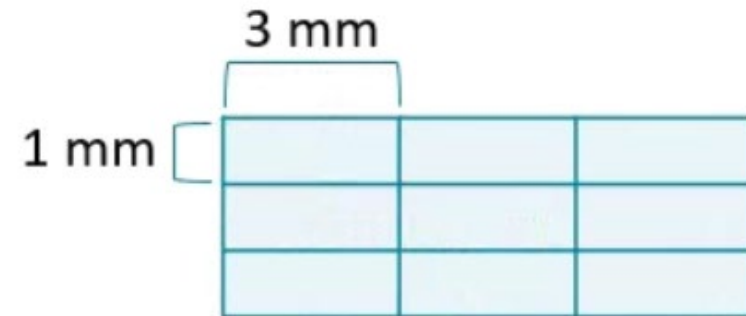
- Many datasets do not have equal sampling rates across all dimensions. In these cases, we will want to **stretch** the pixels along one side to account for the differences.
- Any two dimensions of an array can form an image, and slicing along different axes can provide a useful perspective.
- However, unequal sampling rates can create distorted images.
- Changing the **aspect ratio** can address this by increasing the width of one of the dimensions.

Pixels may adopt any aspect ratio:

4:1

16:9

1:1



Shape: (3, 3), Sampling: (1, 3)

Modifying the aspect ratio (2)

- This results in a properly proportioned image.
- Failing to adjust the aspect would have resulted in a distorted image.

```
In [121]: 1 view_sagittal = vol3[:, :, 300]
          2 d0, d1, d2 = vol3.meta['sampling']
          3 d0, d1, d2
```

```
Out[121]: (2.0, 0.5078125, 0.5078125)
```

```
In [122]: 1 asp1 = d0/d1
          2 asp1
```

```
Out[122]: 3.9384615384615387
```

```
In [124]: 1 plt.imshow(view_sagittal, aspect = asp1)
          2 plt.axis('off')
          3 plt.show()
```



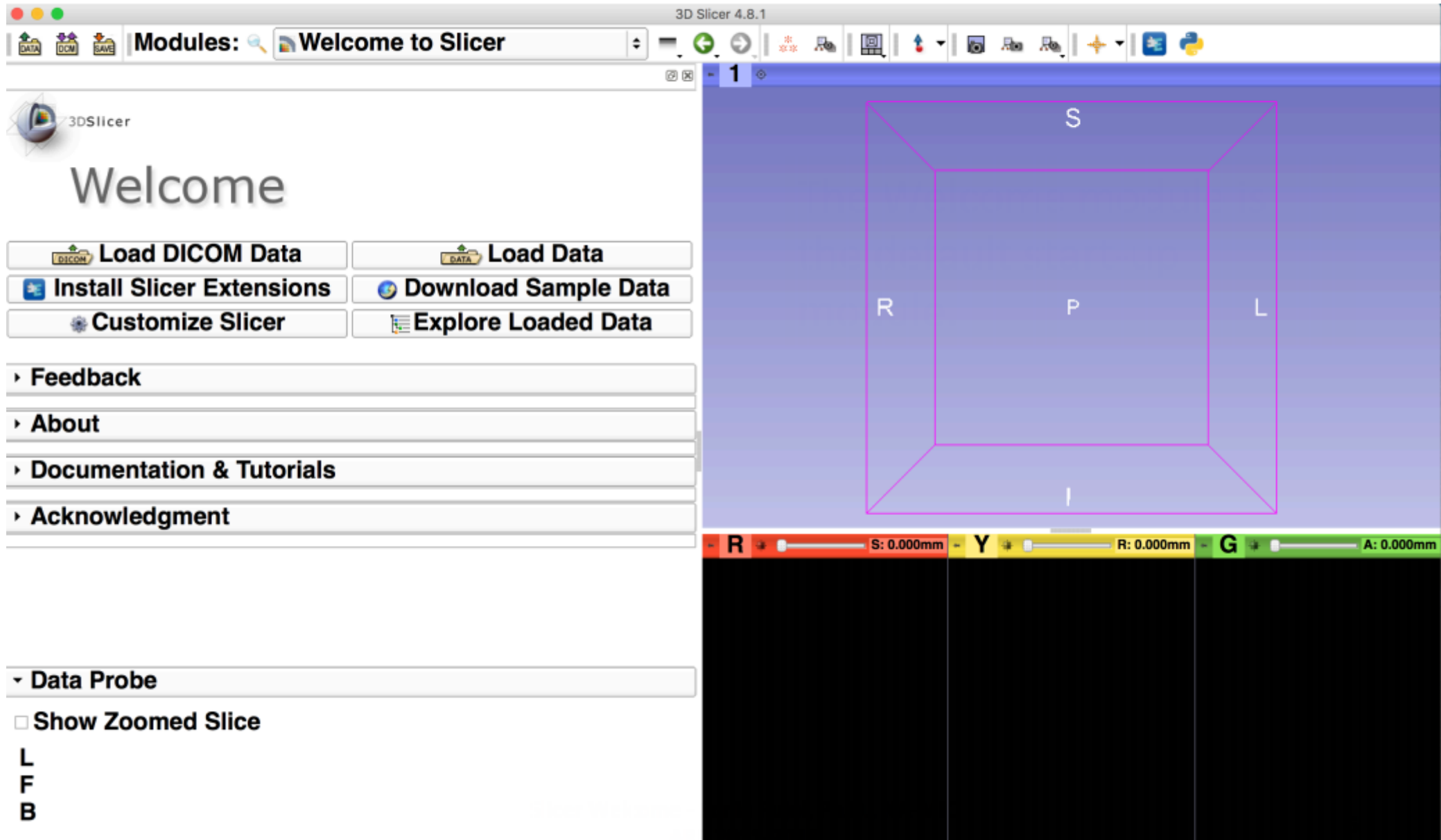


IMAGES IN 3D SLICER

Supported Platforms

- Slicer is a multi-platform software developed and maintained on Mac OSX, Linux and Windows.
- Slicer requires a minimum of 2 GB of RAM and a dedicated graphic accelerator with 64 MB of on-board graphic memory.

3D Slicer version 4.8



Welcome to Slicer

3D Slicer 4.8.1

Modules: Welcome to Slicer

3DSlicer

Welcome

Load DICOM Data Load Data

Install Slicer Extensions Download Sample Data

Customize Slicer Explore Loaded Data

Feedback

About

3D Slicer is a free open source software platform for medical image processing and 3D visualization of image data. This module contains some basic information and useful links to get you started using Slicer. For more information, please visit our website <http://www.slicer.org>.

3D Slicer is distributed under a BSD-style license; for details about the contribution and software license agreement please see the [3D Slicer Software License](#)

Data Probe

☐ Show Zoomed Slice

L
F
B

Each module of Slicer includes a series of tabs, which give access different functionalities.

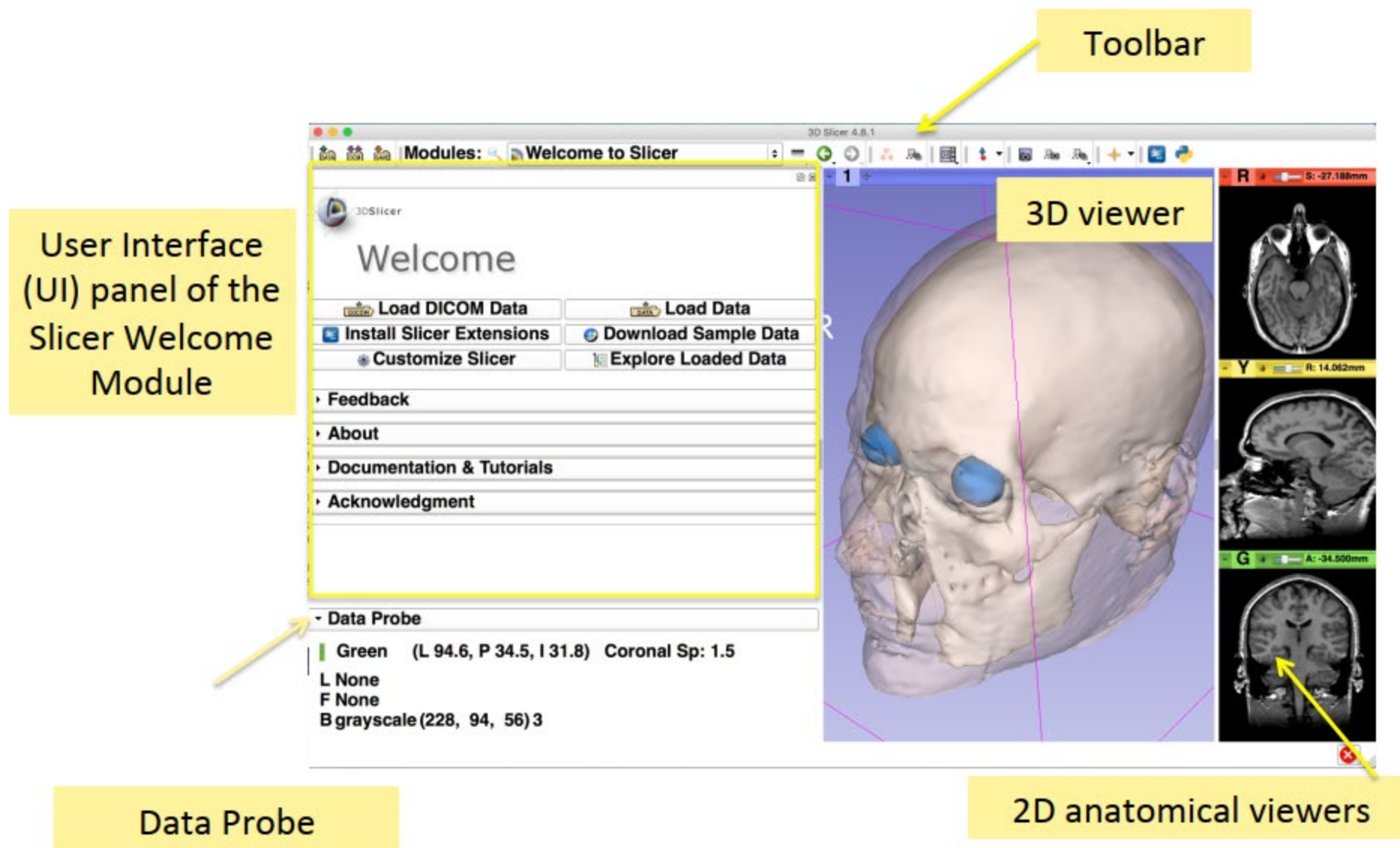
Click on the arrow symbol to display the content of each tab.

R A V

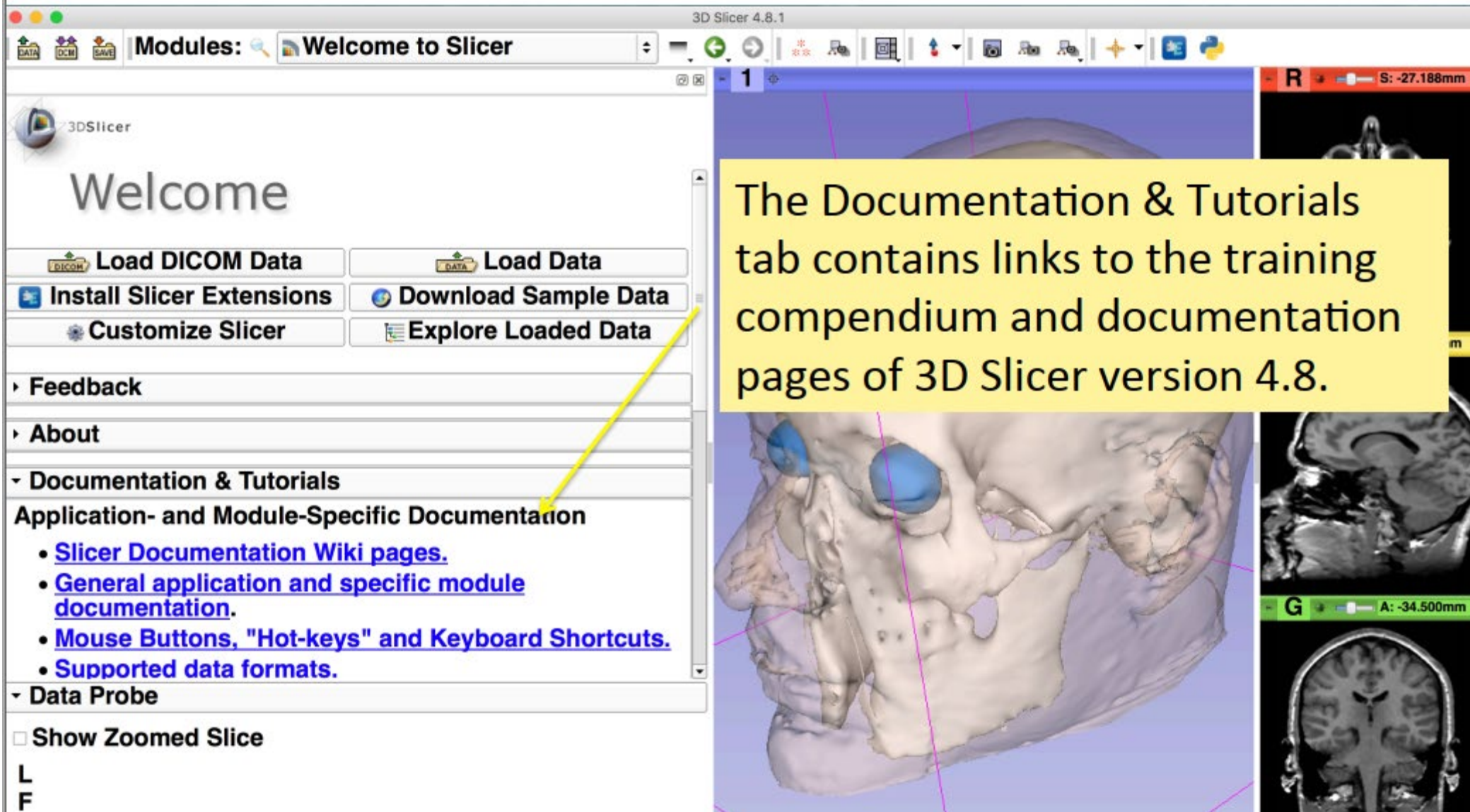
A: 0.000mm

7

Slicer User Interface



Welcome Module



Slicer 4.8 Training & Documentation

Documentation/4.8/Training

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Introduction: Slicer 4.8 Tutorials

- This page contains "How to" tutorials with matched sample data sets. They demonstrate how to (release) to accomplish certain tasks.
- For tutorials for other versions of Slicer, please visit the [Slicer training portal](#).
- For "reference manual" style documentation, please visit the [Slicer 4.8 documentation page](#)
- For questions related to the Slicer4 Training Compendium, please send an e-mail to [Sonia Pujol](#)

- Some of these tutorials are based on older releases of 3D Slicer. The concepts are still useful but features will be different in updated versions.

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 - 4.2 Slicer4 3D Visualization of DICOM Images for Radiology Applications
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 - 8.4 Slicer4 IGT

Documentation/4.8

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Where to start ?

- [Getting started](#)
Quick overview about Slicer
- [Training pages](#)
Information on how to use Slicer 4.8
- [FAQ](#)
Set of common questions/answers
- [Discussion Forum](#) 
The most effective way to get help from the community

How to

- [Report a problem / Create a feature request](#) **NEW**

Slicer Application

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- [Main Application User-Interface](#)
- [Application Settings](#)
- [Extensions Manager](#)
- [Mouse Buttons, "Hot-keys" and Keyboard Shortcuts](#)
- [Recommended Computer Configurations](#)
- [Loading or Saving data and listing of supported data formats.](#)
- [Slicer Lookup Tables](#)

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Developers Corner

- [Information for Software Developers](#) **UPDATED**
Source code, contribute patch, develop modules

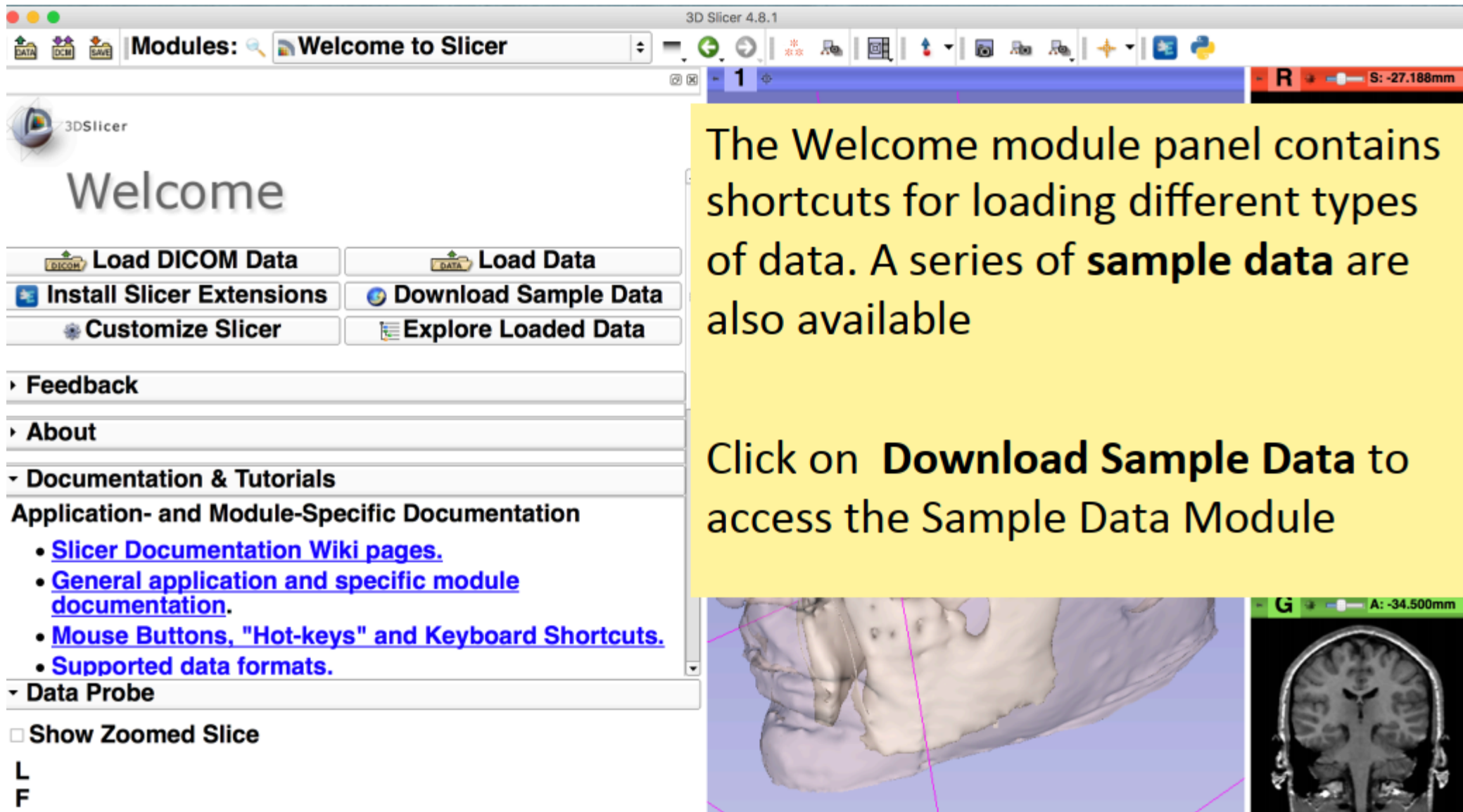
Miscellaneous

- [Documentation guidelines](#)
Slicer user documentation principle and guidelines
- [Visual blog](#)
Set of screenshots showing Slicer in action.
- [Release Notes](#)
Platform specific issues and considerations
- [Announcements & Acknowledgments](#)
- [Registration Library](#) 
Real-life example cases of using the Slicer registration tools, incl. datasets and step-by-step instructions to follow and try yourself.

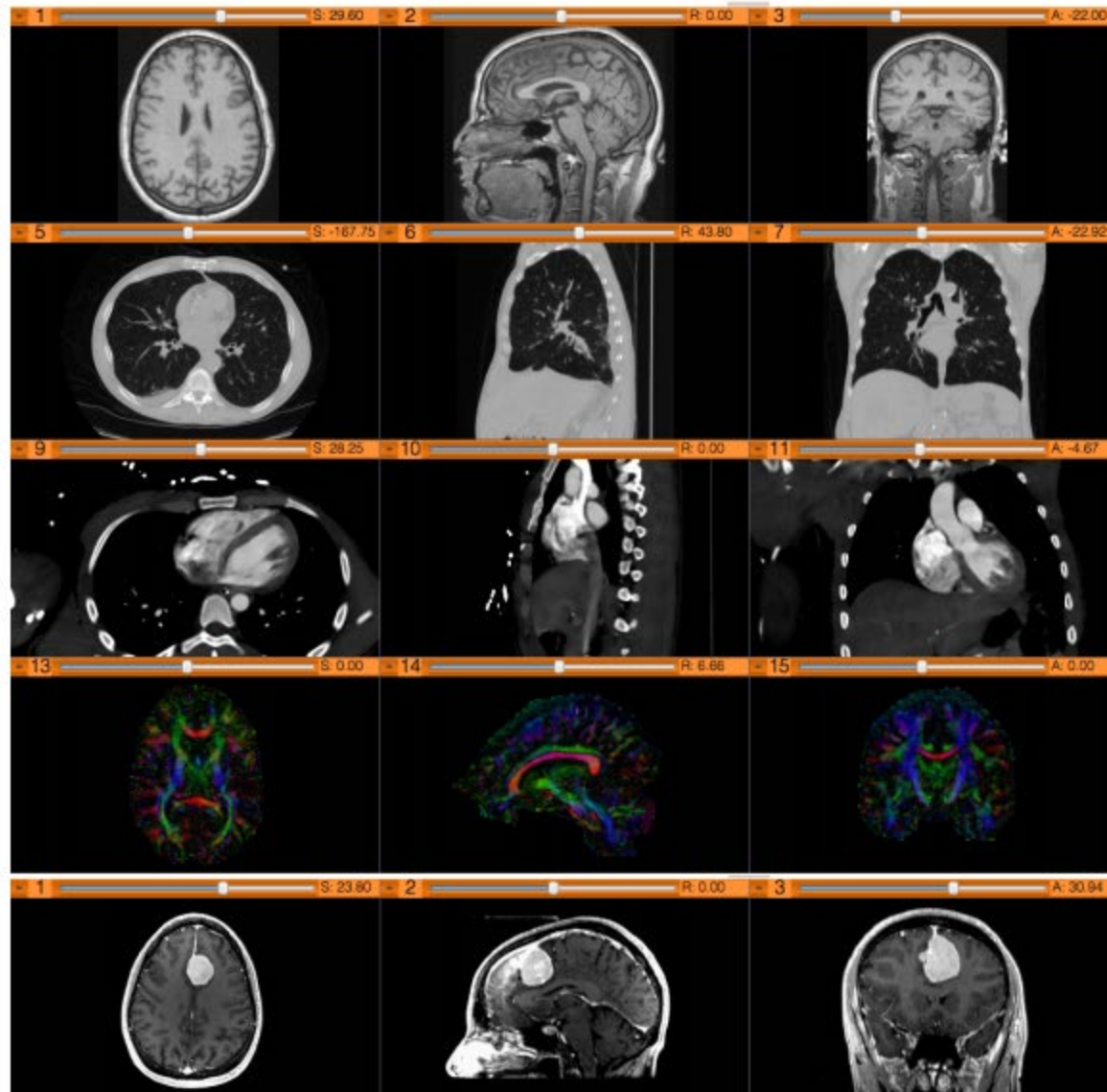
Documentation in other languages

- [Español](#)

Welcome Module



Sample Data



Brain MRI

Chest CT

Cardiac CT

Diffusion Tensor
Imaging (DTI) Dataset

Brain MRI
(tumor patient)

Welcome Module

The screenshot displays the 3D Slicer 4.8.1 software interface. The top toolbar includes icons for file operations (Data, Open, Save), a 'Modules' dropdown menu, and a 'Sample Data' button. Below the toolbar, the left sidebar shows a tree view with categories: 'Help & Acknowledgement', 'Reload & Test', and 'BuiltIn'. Under 'BuiltIn', there are nine data probe thumbnails: MRHead, CT Chest, CTACardio, DTIBrain, MRBrainTumor1, MRBrainTumor2, BaselineVolume, DTIVolume, and DWIVolume. At the bottom of the sidebar, there is a 'Data Probe' section with a checkbox for 'Show Zoomed Slice' and labels 'L', 'F', and 'B'. The main window area is divided into two parts. The top part is a large 2D view showing a purple rectangular frame with labels 'R' (Right), 'P' (Posterior), 'L' (Left), and 'I' (Inferior) at the corners. The bottom part is a 3D view showing three orthogonal slices of a brain MR scan: an axial slice on the left, a sagittal slice in the middle, and a coronal slice on the right. A yellow text box is overlaid on the 2D view, stating: 'The MR scan of the brain appears in the 2D viewers.'

3D Slicer 4.8.1

Modules: Sample Data

1

3DSlicer

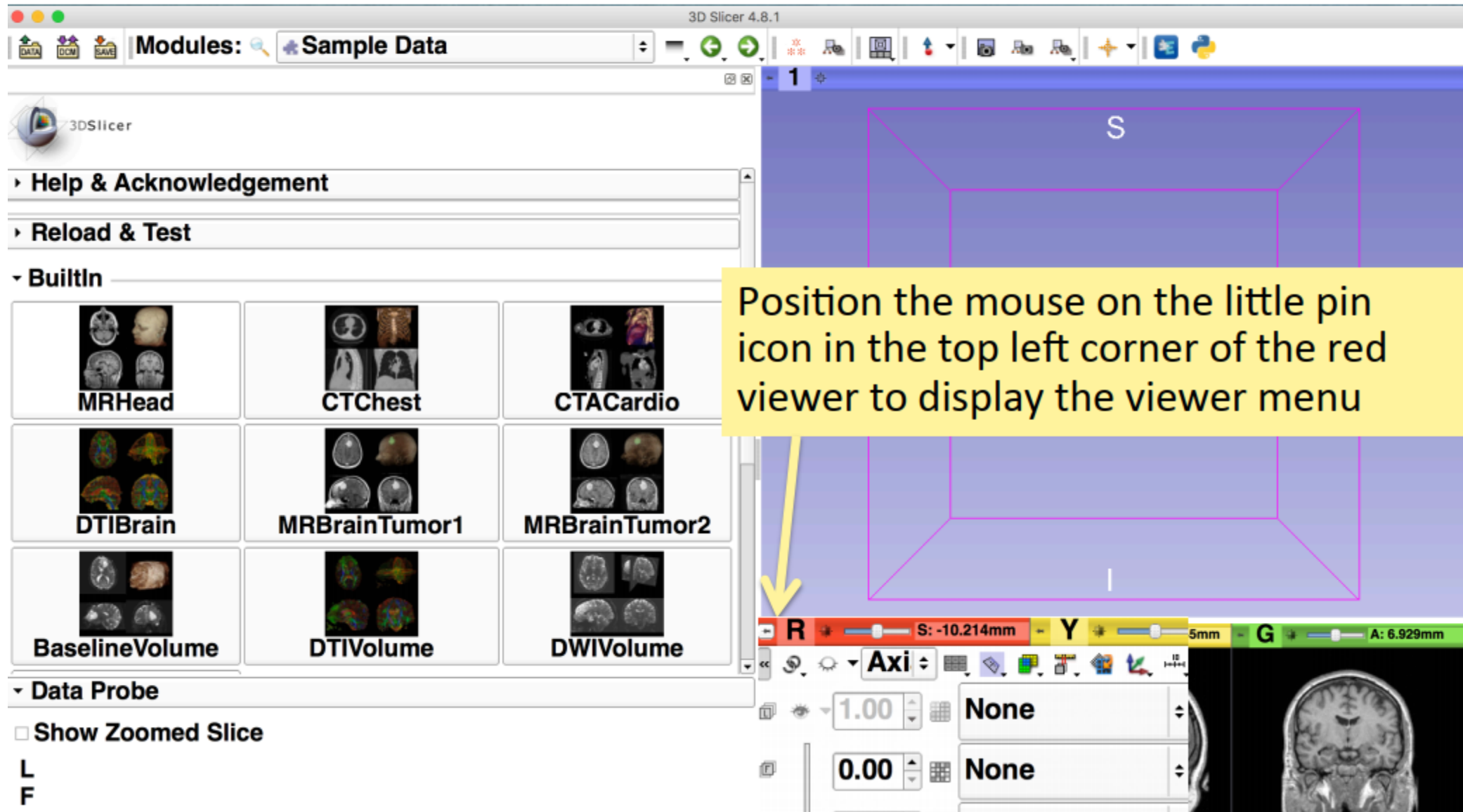
- Help & Acknowledgement
- Reload & Test
- BuiltIn
 - MRHead
 - CT Chest
 - CTACardio
 - DTIBrain
 - MRBrainTumor1
 - MRBrainTumor2
 - BaselineVolume
 - DTIVolume
 - DWIVolume
- Data Probe
 - Show Zoomed Slice
 - L
 - F
 - B

The MR scan of the brain appears in the 2D viewers.

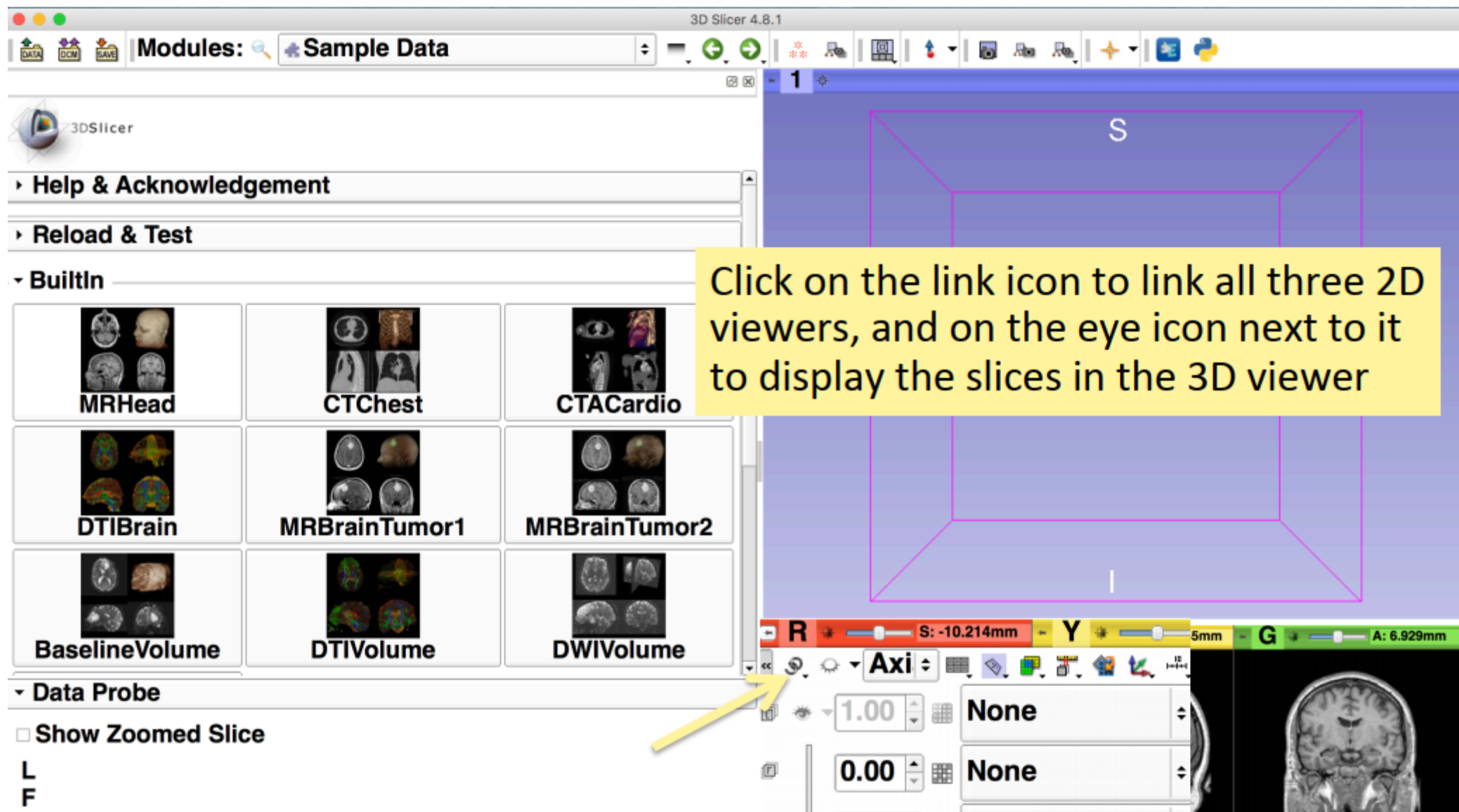
R P L I

R S: -10.214mm Y R: -2.145mm G A: 6.929mm

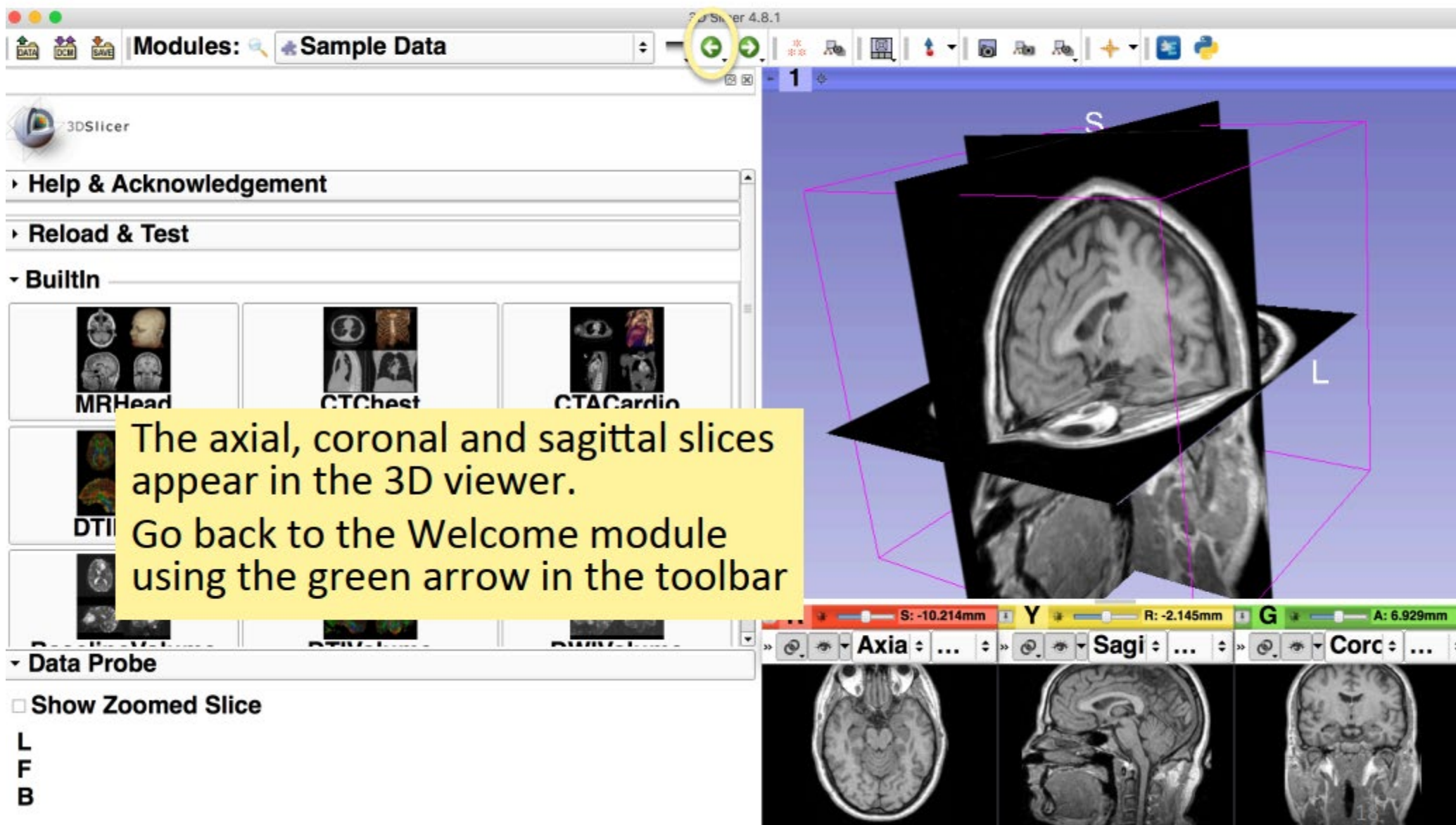
MR Brain Sample Dataset

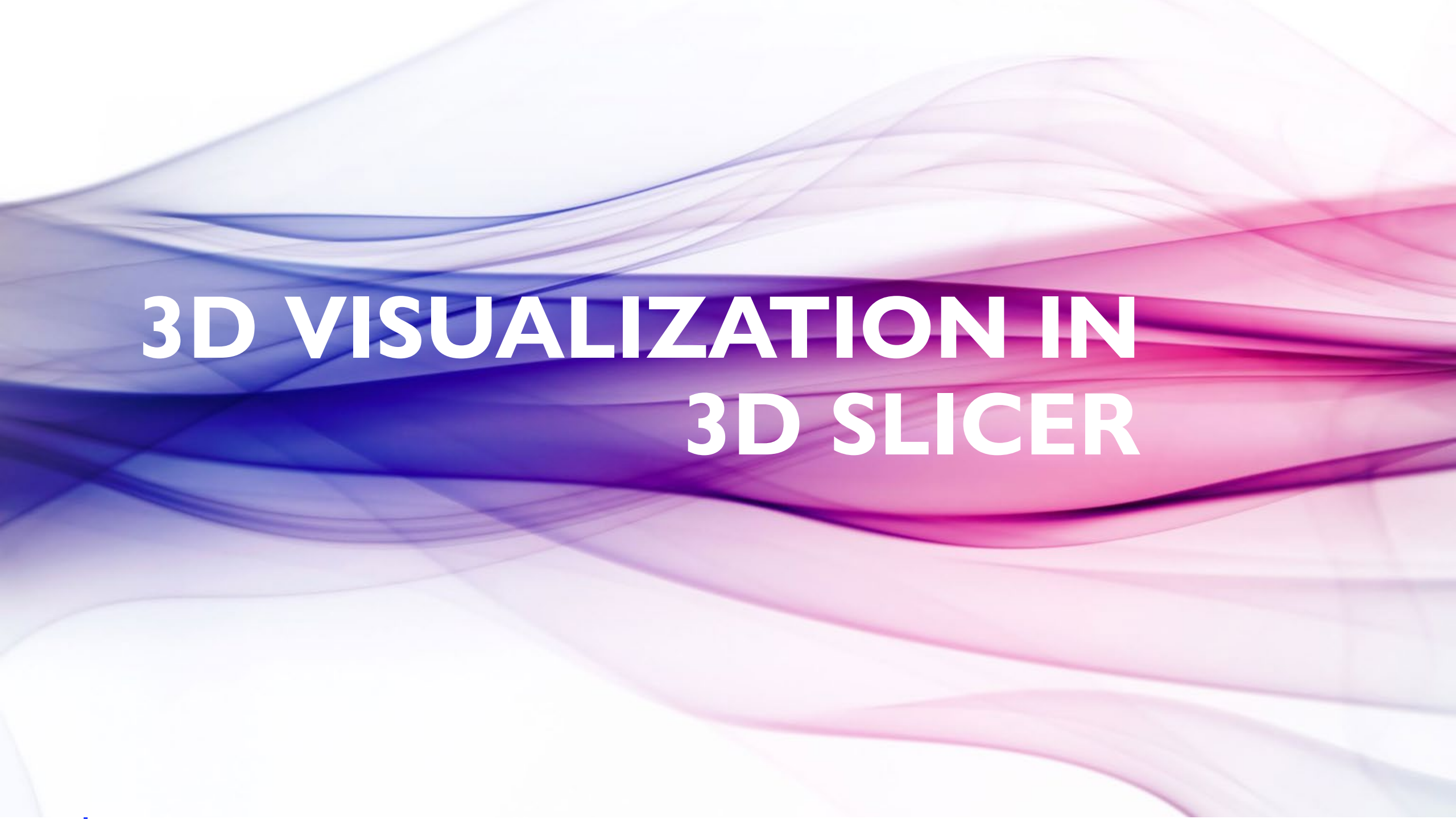


MR Brain Sample Dataset



MR Brain Sample Dataset



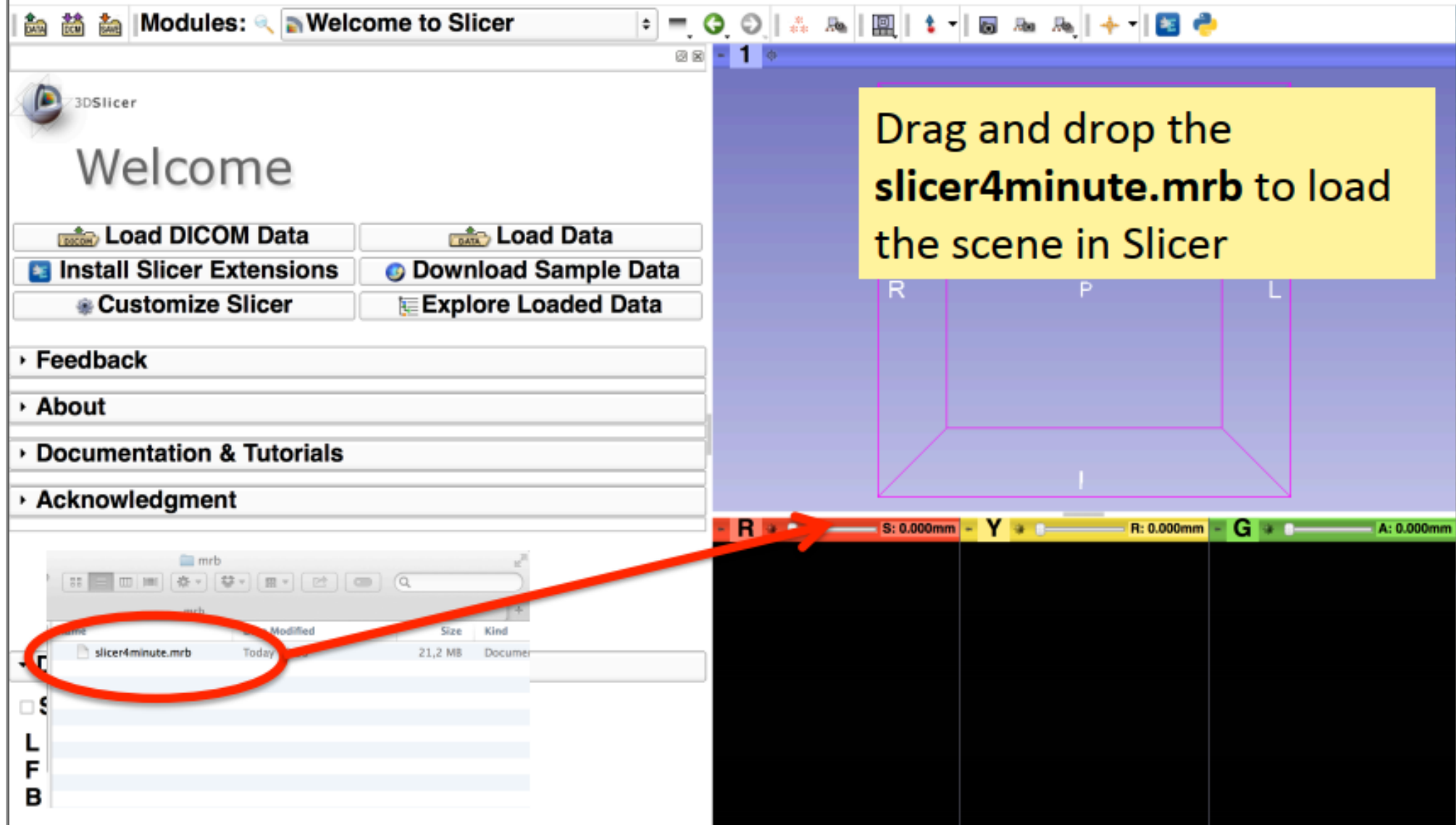


3D VISUALIZATION IN 3D SLICER

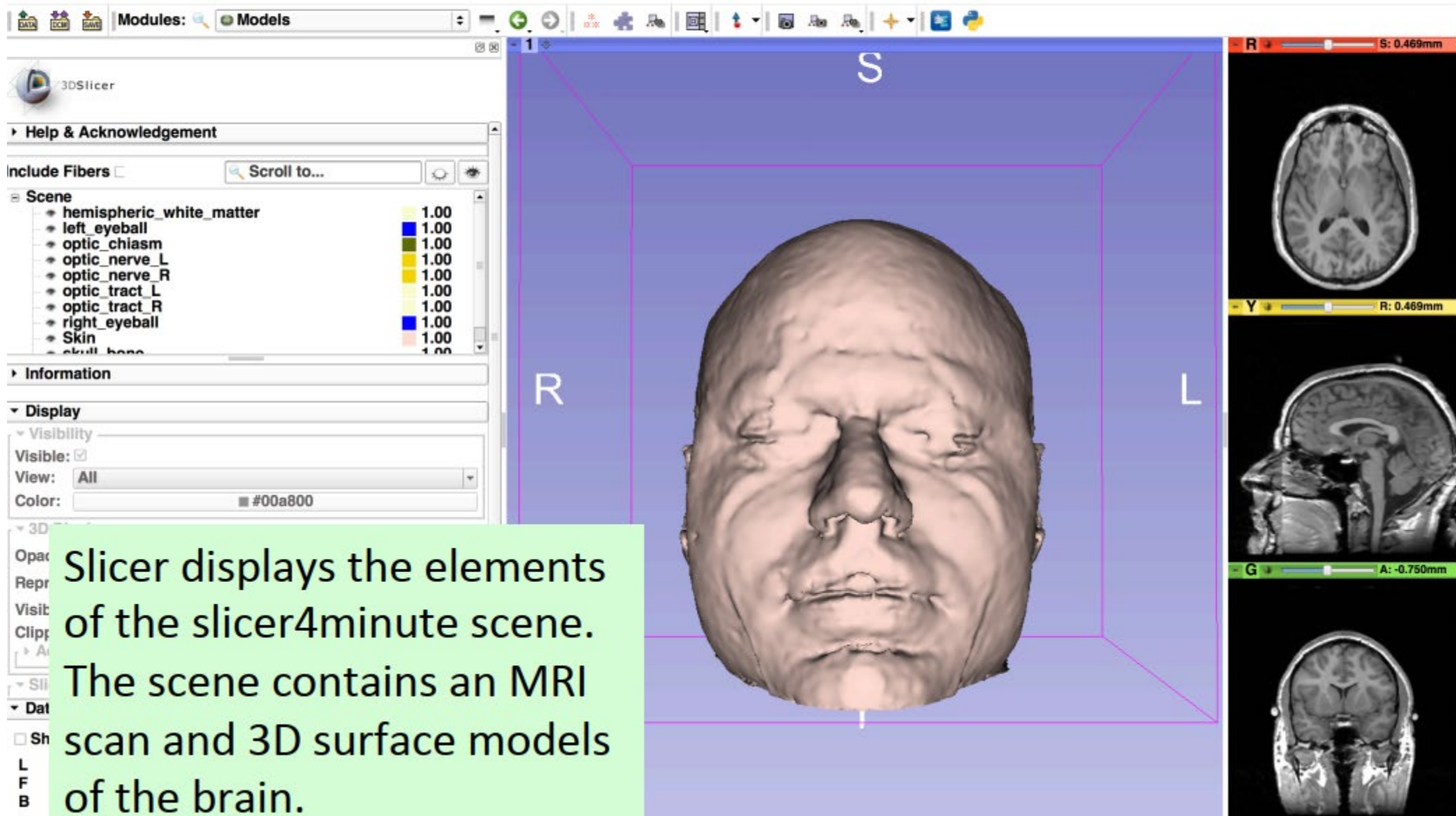
3D Slicer Scene

- A Slicer scene is a **MRML (Medical Reality Modeling Language)** file that contains a list of elements loaded into Slicer (volumes, models, fiducials, transforms, etc.)
- In the following example, we use a scene 'Slicer4minute.mrml' composed of an MRI scan and 3D models of the head.
- The scene file and datasets have been saved as a **MRB** (Medical Reality Bundle) file.
- The MRB file format is Slicer's archive file format.

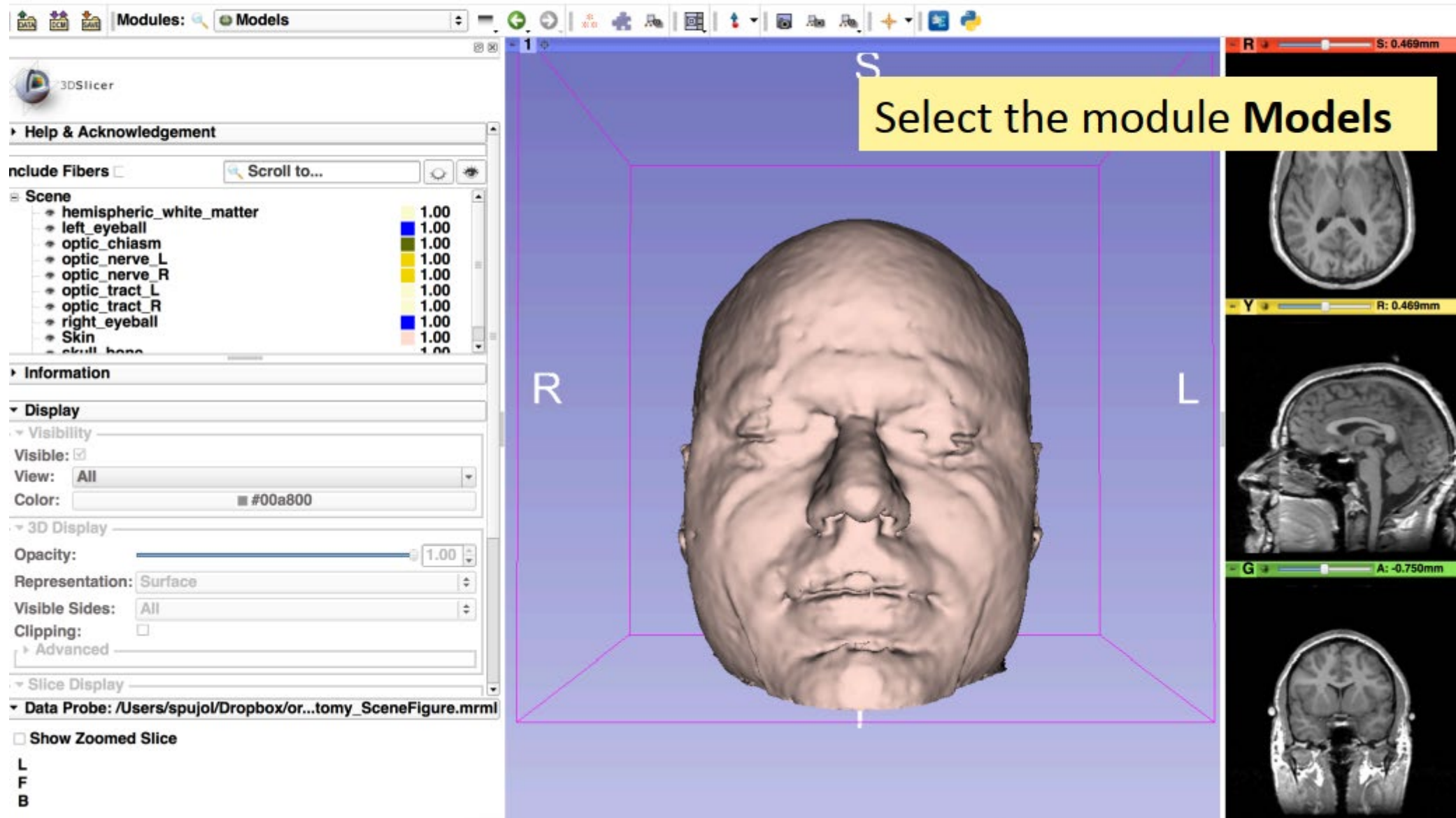
Loading the Slicer4minute dataset



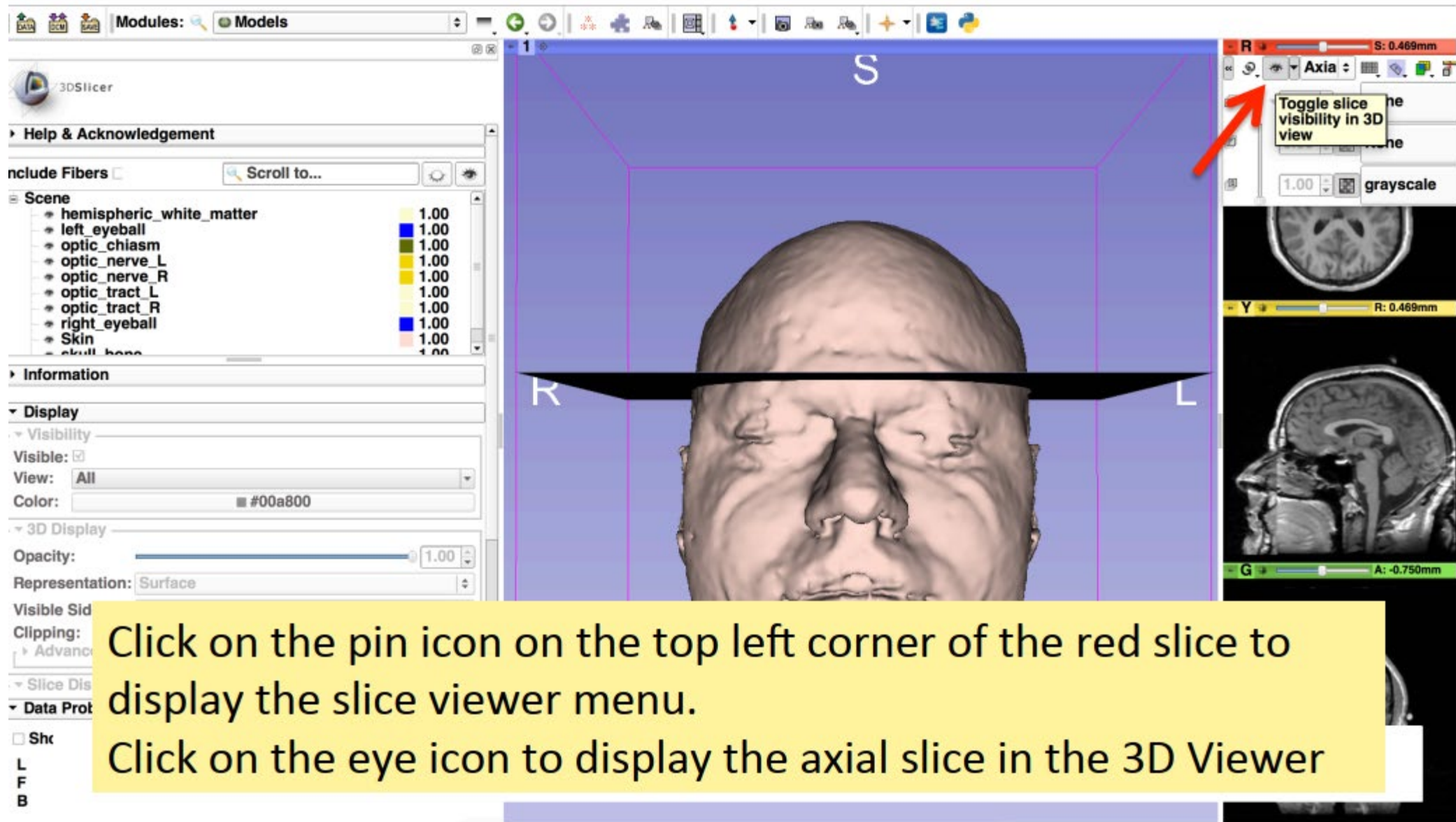
Slicer4minute Scene



3D Visualization

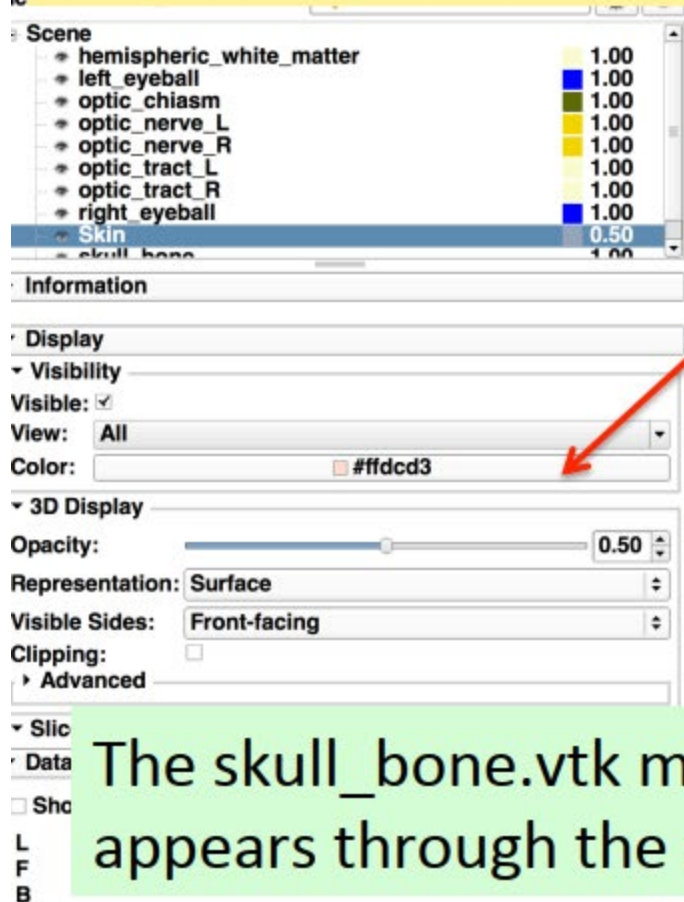


3D Visualization

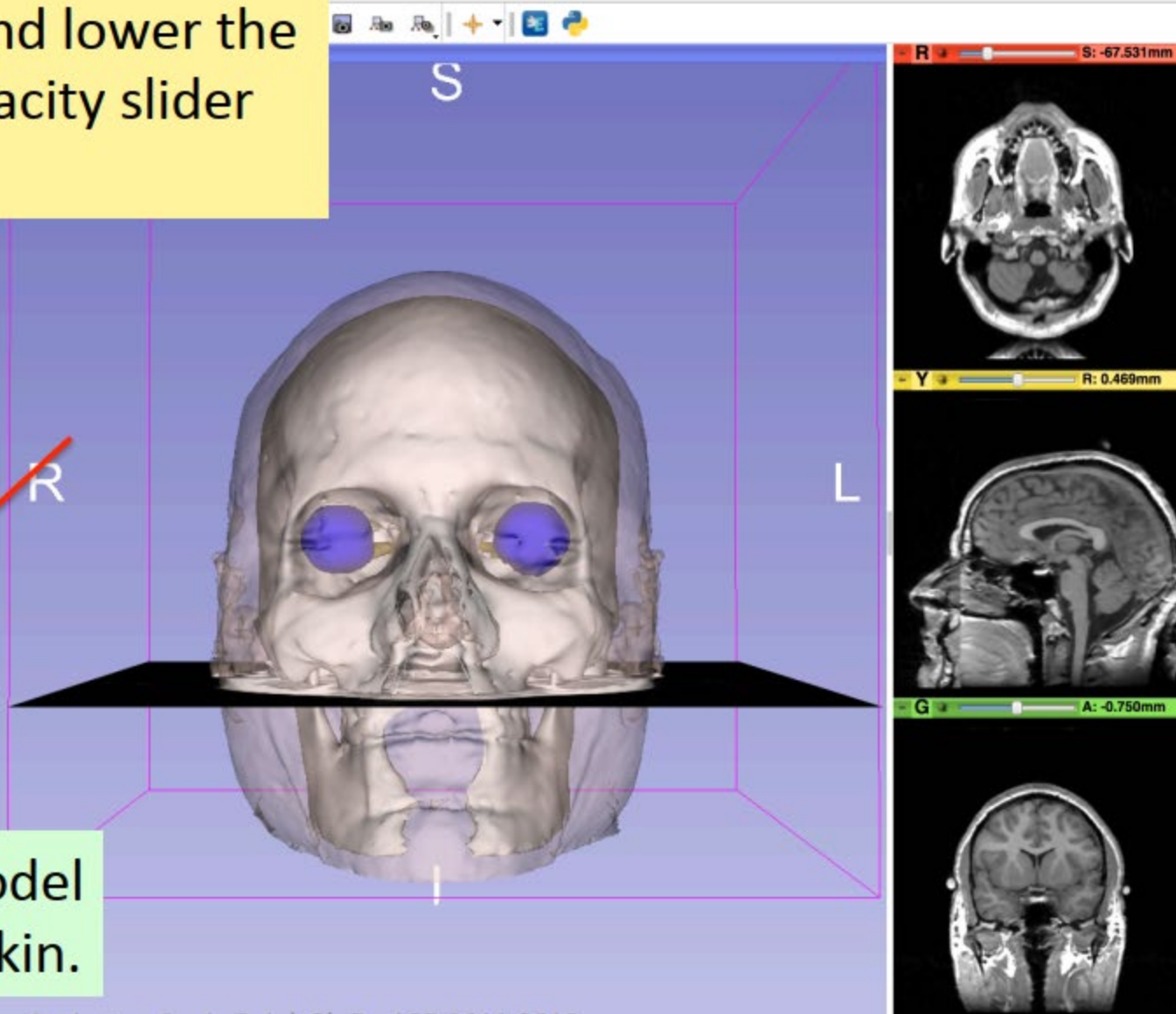


3D Visualization

Select the Skin model and lower the its opacity using the Opacity slider in the 3D Display tab



The skull_bone.vtk model appears through the skin.



3D Visualization

Position the mouse in the 3D viewer, click on the left-mouse button to drag and rotate the model. Click on the right-mouse button to zoom in and out

Information

Display

Visibility

/isible: ☐

/iew: All

Color: #ffdcd3

3D Display

Opacity: 0.50

Representation: Surface

/isible Sides: Front-facing

Clipping: ☐

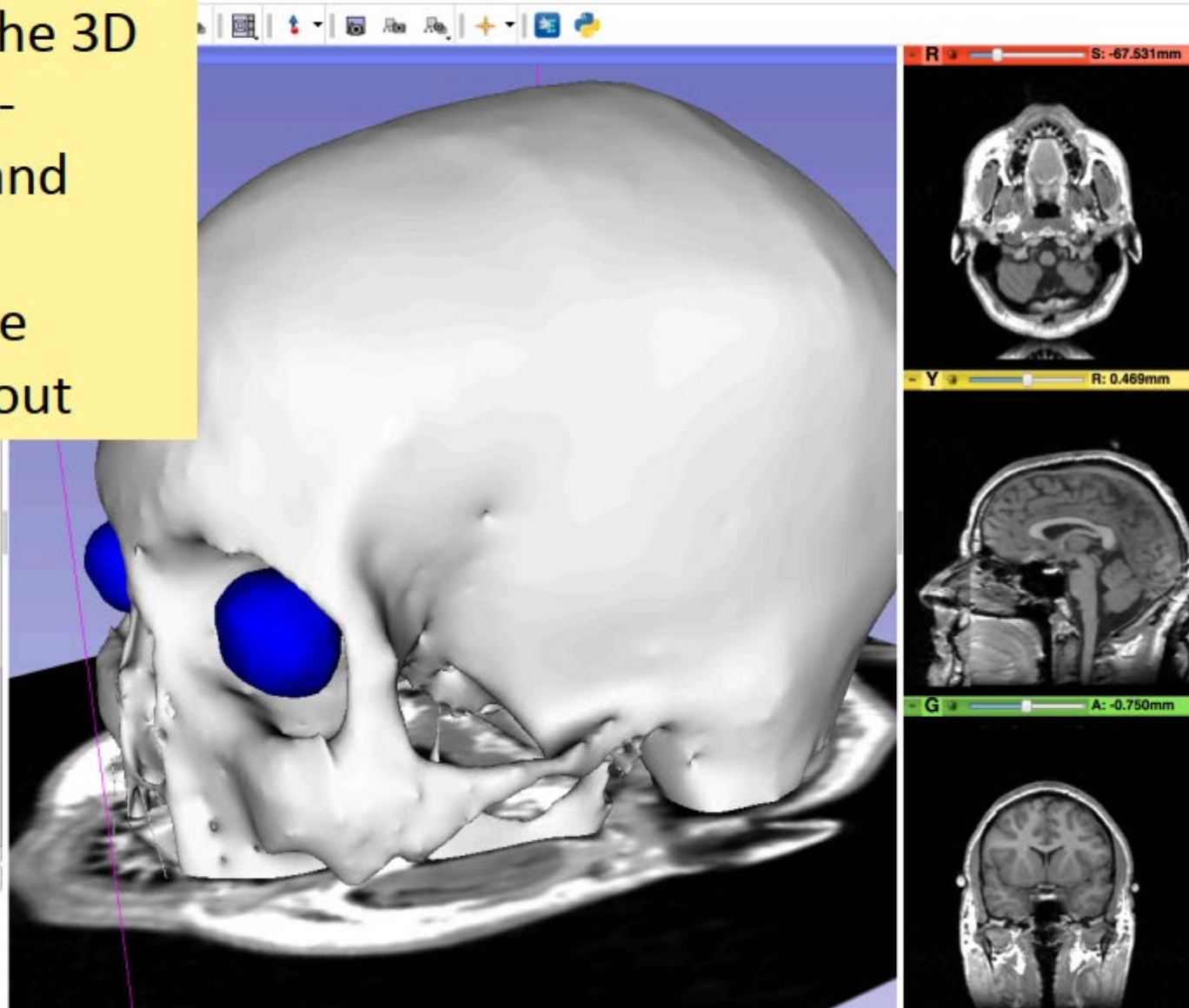
Advanced

Slice Display

Data Probe: /Users/spujo/Dropbox/or...tomy_SceneFigure.mrml

Show Zoomed Slice

L
F
B



Anatomical Views

Click on the pin icons in the top left corner of the red and green viewer to display the slice viewer menu

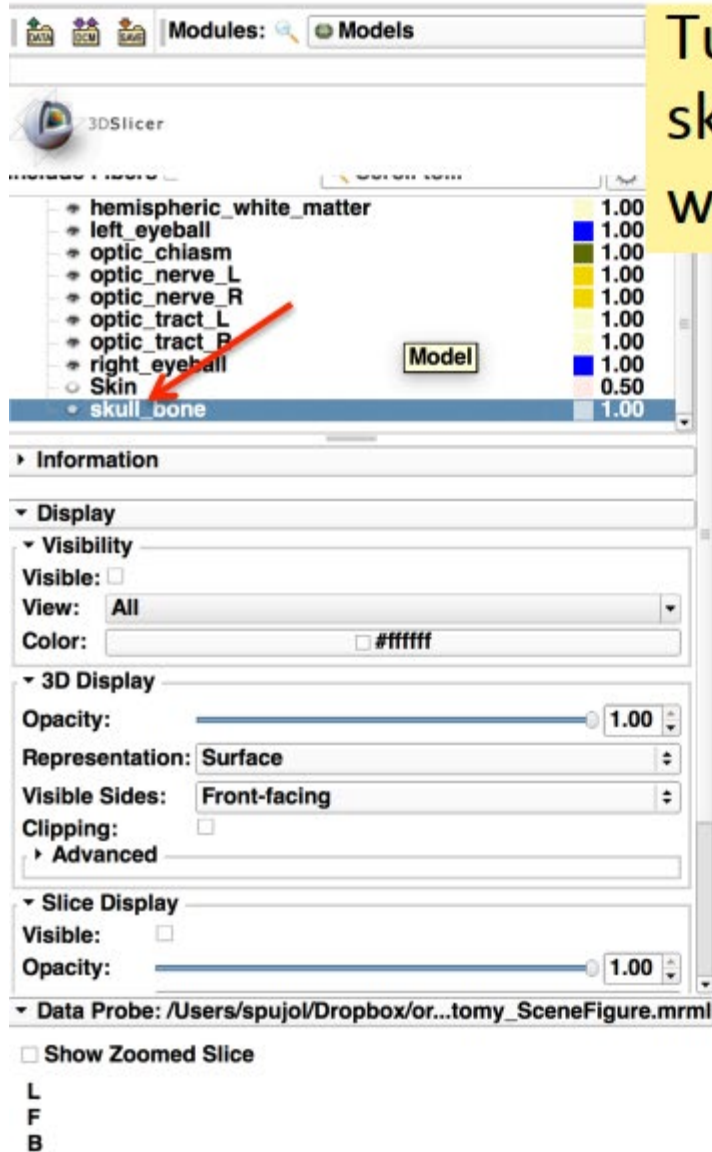
Click on the eye icon to display the axial and coronal slice in the 3D viewer

Visible Sides: Front-facing
Clipping: ☐
Advanced
Slice Display
Data Probe: /Users/spujol/Dropbox/or...tomy_SceneFigure.mrm
☐ Show Zoomed Slice
L
F
B

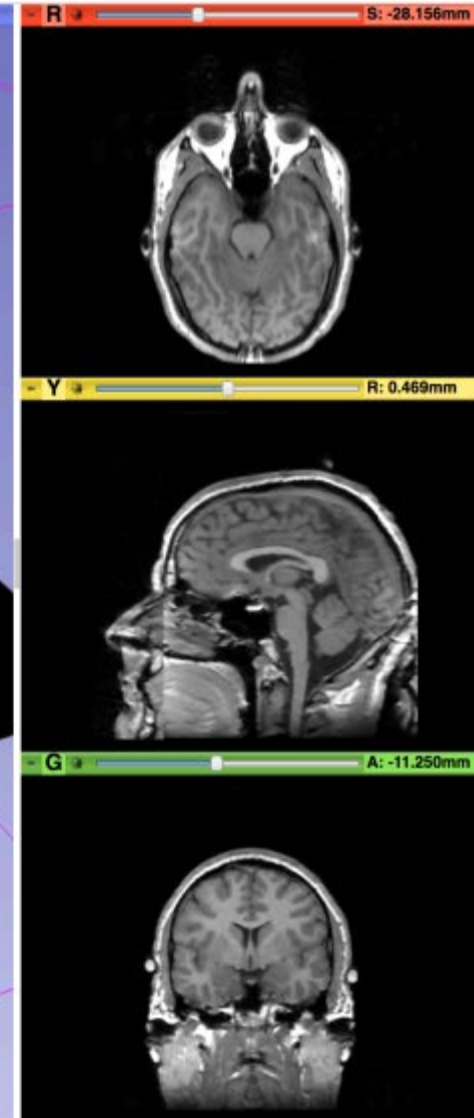
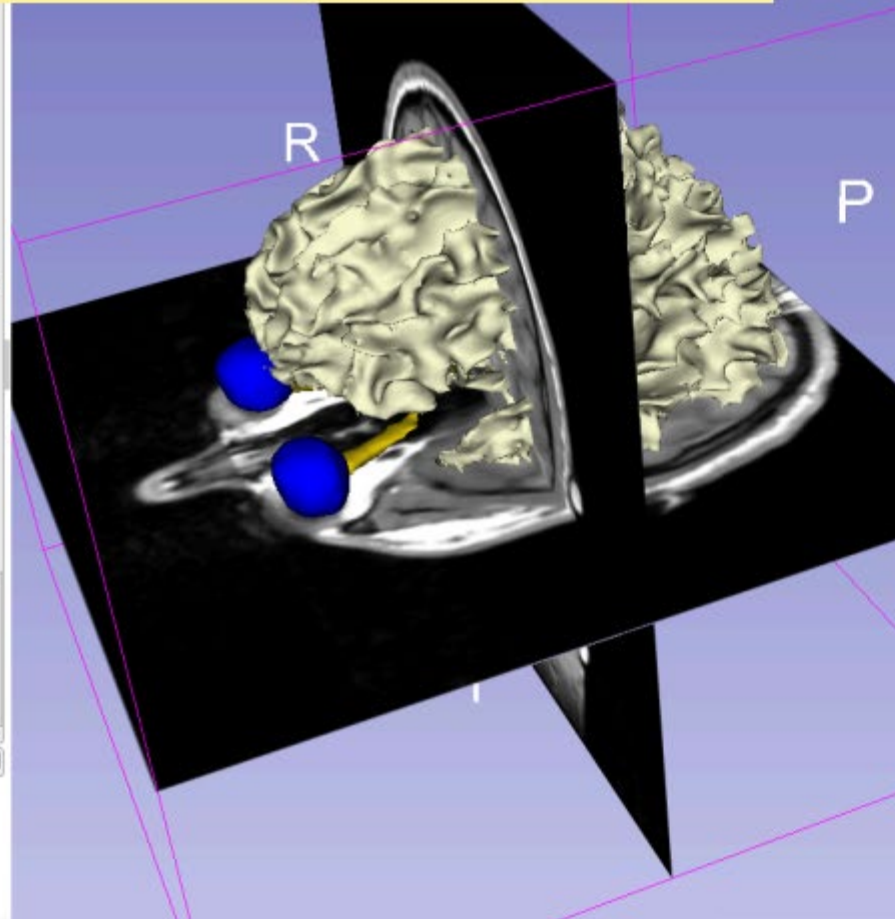
R S: -67.531mm
Axial
1.00 None
0.00 None
1.00 grayscale

G A: -0.750mm
Coro
1.00 None
0.00 None
1.00 grayscale

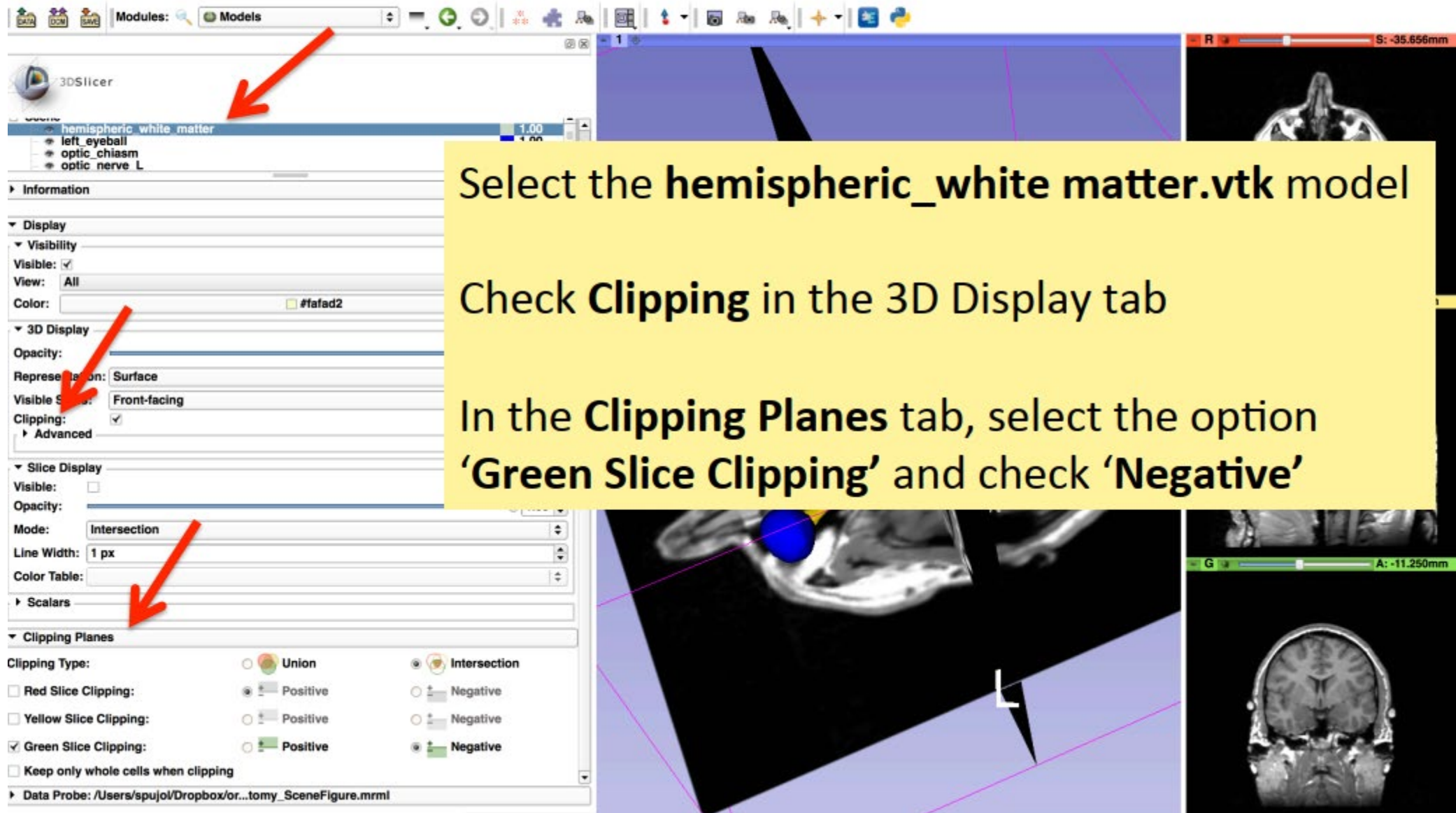
3D Visualization



Turn off the visibility of the skull to display the brain white matter model



3D Visualization



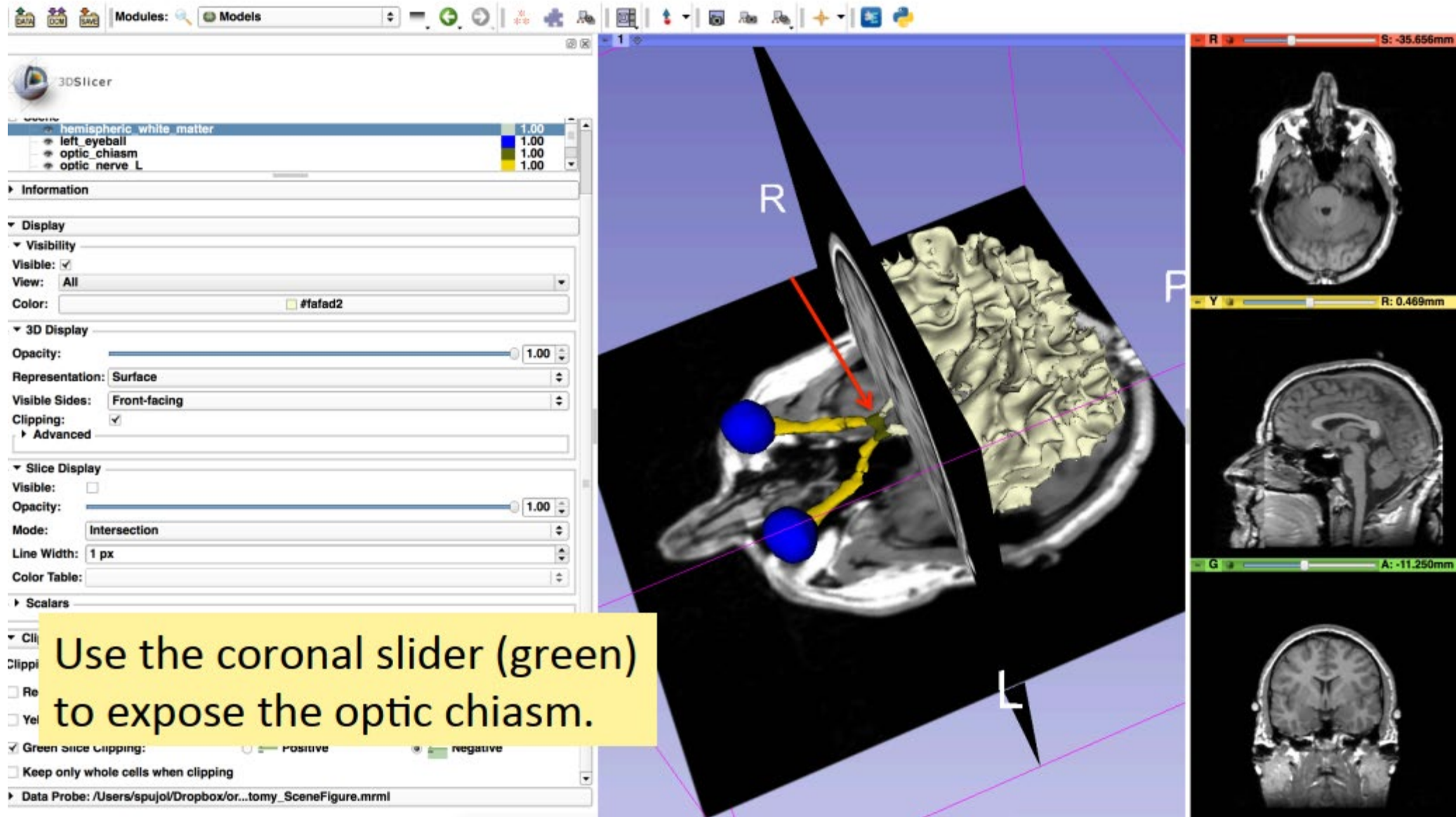
The screenshot shows the 3D Slicer software interface. The left sidebar contains a 'Models' list with 'hemispheric white matter' selected. Below it, the 'Information' panel shows the '3D Display' tab. In this tab, 'Clipping' is checked. The 'Clipping Planes' section at the bottom shows 'Green Slice Clipping' selected with 'Negative' checked. The main 3D view displays a brain model with a blue sphere and a green slice. The right sidebar shows the 'Clipping Planes' tab with 'Green Slice Clipping' selected and 'Negative' checked. The bottom right corner shows a 'Data Probe' window with coordinates.

Select the **hemispheric_white matter.vtk** model

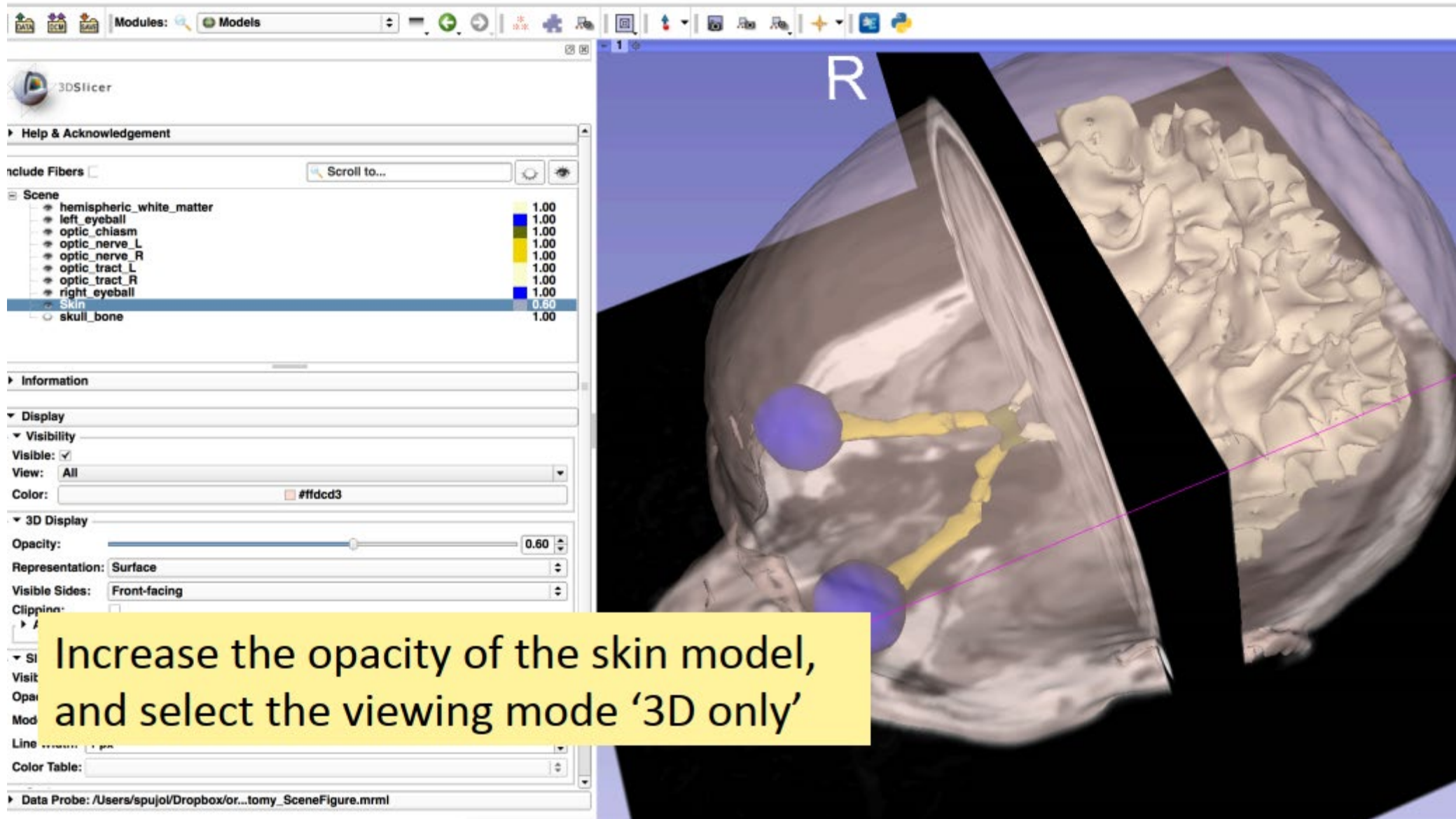
Check **Clipping** in the 3D Display tab

In the **Clipping Planes** tab, select the option '**Green Slice Clipping**' and check '**Negative**'

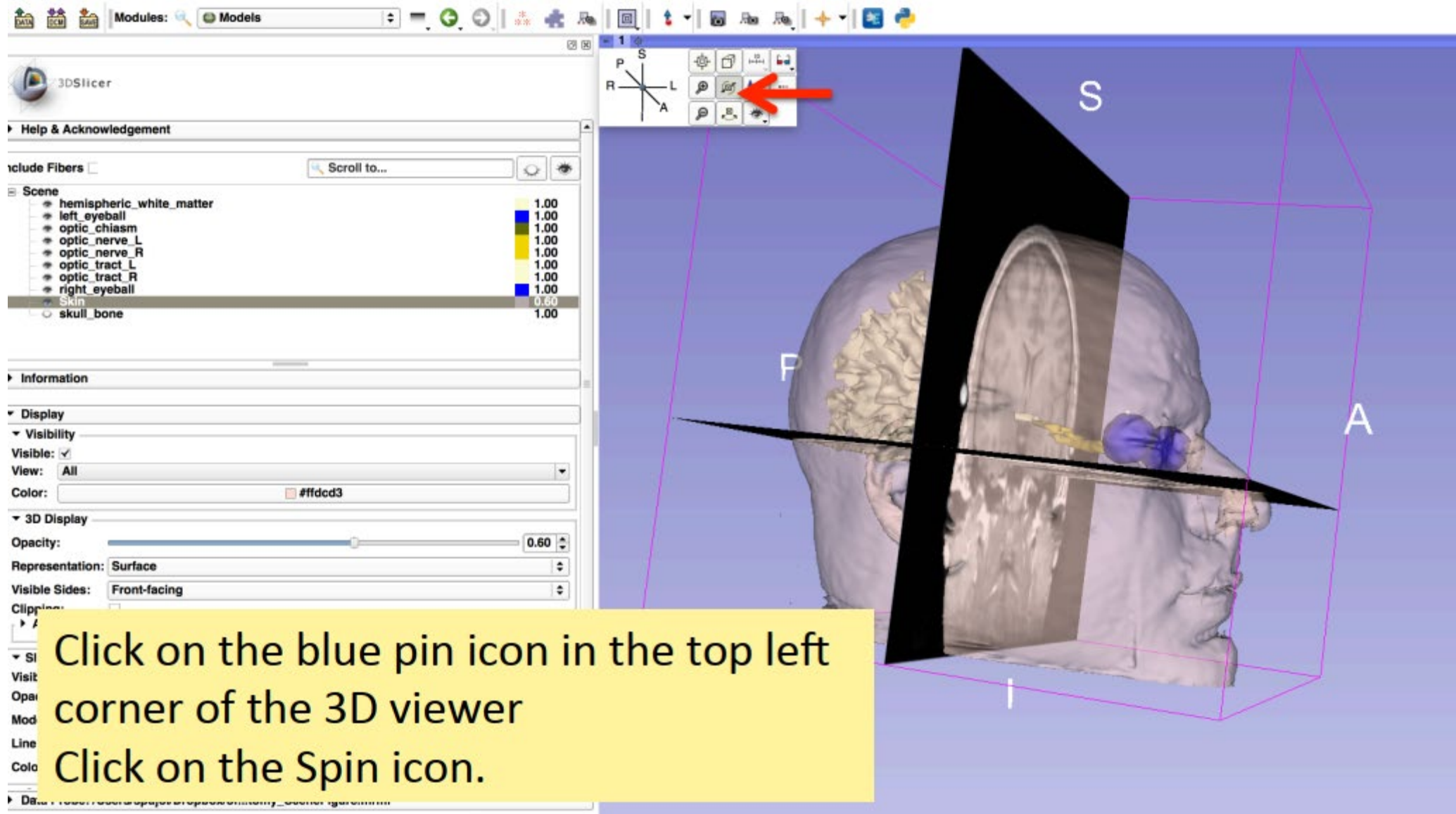
3D Visualization



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Acknowledgments



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Neuroimage Analysis Center

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